

Statistical Machine Learning - HW 3

Problems 1-4: Chapter 5

- Exercise #1, #3, #5, #8 from Chapter 5

Problem 5: Chapter 6

Show that the ridge regression estimate is the maximizer of the (log) posterior distribution, under a Gaussian prior $\beta \sim N(0, \tau^2 \mathbf{I})$, and Gaussian sampling model $\mathbf{y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I})$. Find the relationship between the regularization parameter λ in the ridge formula, and the variances τ^2 and σ^2 .